Lecture 1: Learning and overfitting

Machine learning fits patterns from training examples.

Overfitting occurs when a model learns noise in the training data.

An overfit model performs well on training data but poorly on unseen data.

Regularization and simpler models can reduce overfitting.

Lecture 2: Evaluating models

Training data is used to fit model parameters.

Validation data is used to choose hyperparameters and compare models.

Test data is reserved for the final evaluation of generalization.

Do not tune hyperparameters on the test set; this can bias evaluation.